Pointers Master Practice Sheet

Easy

1. Pointer Basics

Declare a pointer and assign it the address of an integer variable. Print the value and the address using the pointer.

2. Swap using Pointers

Write a function to swap two numbers using pointers.

Sample Input: a = 10, b = 20

Sample Output: a = 20, b = 10

3. Array Access via Pointers

Input an array and print all its elements using pointer notation.

4. Pointer Arithmetic

Demonstrate pointer increment and how to access elements via pointer arithmetic.

5. Null Pointers

Declare a null pointer and check if it's null before accessing or dereferencing it.

6. Function with Pointer Parameter

Write a function that takes a pointer to int and updates its value.

Medium

1. Pointer to Pointer

Demonstrate usage of pointer to a pointer (int **pp).

2. Dynamic Memory Allocation

Allocate an array dynamically using pointers, take input, and compute the sum.

3. Reverse Array using Pointers

Reverse an array in-place using pointer manipulation.

4. Pass Array to Function

Write a function that takes a pointer to an array and its size, and prints the elements.

5. String Length (Pointer Version)

Input a string and compute its length using only pointers (no strlen).

6. Change String Case

Convert lowercase letters to uppercase using pointer traversal.

7. Compare Two Strings

Implement your own version of strcmp() using pointers.

8. Count Vowels using Pointers

Input a string and count how many vowels are present using pointer traversal.

9. Pointer with Structures

Define a struct `Student`. Create its object using pointer and access/update its fields.

Hard

1. Pointer to Array of Pointers

Manage multiple strings using an array of character pointers.

2. Dynamic 2D Array with Double Pointer

Dynamically allocate a 2D array using double pointers and take input/output.

3. Linked List Node with Pointer

Create a node using struct and pointer, input value, and print it.

4. Pointer-based Matrix Multiplication

Dynamically input two matrices and multiply them using only pointers.

5. Pointer vs Reference (C++)

Write a program comparing pointer behavior vs reference behavior in C++.

Extended Challenges

1. Simulate strcpy() using Pointers

Implement your own version of strcpy() using character pointers.

2. Reverse a Linked List (Pointer Based)

Reverse a singly linked list using pointer manipulation.

3. Check Palindrome with Pointers

Check if a given string is a palindrome using pointer traversal.

4. Memory Leak Demonstration

Create a program that causes a memory leak, and then fix it.

5. Custom strdup()

Create a function like strdup() using dynamic allocation and pointers.